Response to Reviewer FBhK

BioCoLoop, submission 5046 | Revision dated 10 November 2026. Prepared for author review; not posted to OpenReview.

Thank you for identifying two questions that need separate evidence: whether history improves search at fixed access and budget, and whether development-selected changes generalize. We have added the requested random/direct controls, a fresh three-seed history-representation experiment, and a direct diagnosis of the validation failures. The original results remain visible in consolidated tables.

1. Same ten laboratories, configuration library and search budget (Weakness 1; Q1)

Section 4.7 and Appendix E.1-E.2 now compare random search without replacement, GP-EI, and direct/history-guided Qwen and Luna using the same ten training laboratories, ten development panels, twelve configurations, initialization and checkpoint-selection rule. Each search receives six proposal slots beyond D00, with 100 rounds per fit and a 700-round cap. Random search receives the loop's search budget, not the fixed model's single-fit budget.

We verified all 60 task/configuration fits. Enumerating all 462 six-design subsets gives the exact expected outcome of random search conditional on this cache, with selection based only on development evidence. This is not 462 independent training repetitions. Shared-first-proposal and realized-fit-count controls additionally account for failed LLM proposals; for example, four-valid-proposal random search reaches 26.30 on Tahoe versus Qwen's 25.37 under the same realized fit count.

The comparison identifies a positive case without implying a universal gain: Norman/Luna history reaches 23.11 Top-1, versus 22.30 for direct/GP-EI and 22.66 for random search. DTI favors direct Luna/GP-EI (94.60 versus history's 94.27); proteomics ties throughout. On VCC, direct/history and GP-EI retain the fixed reference at 31.50, while random search averages 27.53. We report all five endpoints and all controls.

The main access-scenario table answers a different question: what a laboratory gains by participating in collaborative research instead of developing a model with only its own measurements. Broader access adds training data, development evidence and local computation. We now distinguish this workflow comparison explicitly from the same-access search table; it is not evidence for history alone.

2. Fresh history-representation experiment

Appendix E.6 adds 48 newly trained design/seed fits and new paired Qwen2.5-7B-Instruct search runs. Four study-cell contexts from Replogle and Nadig provide 1,208 training genes, 414 development genes and 406 held-out genes, with gene-disjoint partitions. This is a separate mechanism benchmark, not a replacement score for the original cell endpoints.

Direct, raw-history and structured-history search share the first proposal and six-request budget. Raw and structured history observe the same queried development facts; structured cards explicitly organize configuration changes, reference comparisons and expected-versus-observed outcomes. Direct, structured history, random search and GP-EI each use seven full fits, 16,940 optimizer steps and 532,840 training-example exposures per seed. We report additional LLM costs separately. Raw history has three invalid slots across the three runs, which remain charged rather than silently replaced.

Mean held-out Top-1 is 22.625 for direct, 22.625 for raw history, 22.888 for structured history, 22.668 for random search and 23.152 for GP-EI. Structured versus direct yields two gains and one tie, averaging +0.263 points; its paired gene-cluster 95% interval is [-1.022, 1.568]. Thus the revised representation changes candidate selection and has a positive point estimate, but does not establish stable superiority over simpler search. These complete controls define the scope of our history claim.

The separate 10.84-point Norman result compares training allocation with uniform fitting, not LLM history. New promotion controls show that random promotion reaches 22.63 versus 22.32 for the trajectory policy. We retain that comparison and do not use it as evidence of a unique history mechanism.

3. What explains the development-test divergence? (Weakness 2; Q2)

We reconstructed all 14 distinct checkpoints selected in the extended VCC/Tahoe searches, re-evaluated their 140 development panels, and independently reproduced all 80 archived prefix test scores. There is no aggregation mismatch in these cases: each VCC panel contains two development interventions and each Tahoe panel six, so equal-panel macro Top-1 is exactly the pooled intervention-macro score, even when query counts differ.

The development population is small: 20 VCC interventions and 60 Tahoe compounds. Qwen/VCC's +1.17 development points accompanies -11.89 test points; Tahoe's +5.00 development points represents three additional correct development compounds, while test performance decreases by 11.94 points. To test dependence on the option sets, we evaluated the same initial/final checkpoints on 20 paired development-roster redraws, without training or reselection. Their mean advantages become -1.34 and -1.33 points, positive in only 6/20 and 7/20 redraws. This is direct evidence of roster sensitivity, not proof that one particular cause explains all failures. Finite-sample selection instability is a known model-selection concern Cawley and Talbot, 2010.

The larger follow-up retains every development/test prefix for all methods and three seeds. All final search outcomes exceed their fully fitted D00 references, so the earlier large terminal decreases do not recur. A small interim reversal remains: in seed 62, raw and structured history select D05 at slot 2, improving development by 0.057 points but reducing test by 0.257; later selections recover. A context-specific reversal also remains on K562. Because the data, response panel and candidate library differ, we do not attribute this change to feedback formatting alone or claim that selection error is eliminated. The follow-up supplies a new gene-disjoint study, not repeated regroupings of the original VCC/Tahoe partitions.

4. Source-defined panel feedback and remaining selection instability

Appendix E.8 repairs and audits a separate implementation issue in the older source-defined protein/cell adapters. Ten target development panels were averaged inside one worker, so the outer worst-client field equalled the mean instead of the true worst panel. This did not change the primary selection score. A new adapter/reducer/projection version preserves explicit panel labels, primary/loss and dispersion, gives raw/structured cards identical observed facts, and keeps direct free of outcome feedback.

We froze the audit before new development inference and reconstructed every saved best checkpoint: two tasks, twelve designs and three seeds, or 72 checkpoints and 720 labelled panel scores. Primary/loss reproduce exactly; mean and true worst-panel scores differ in all 72 cases. Eight new paired ten-round CPU canaries on D00/D07 preserve training losses, selected rounds, selected weights and final training states. These are engineering checks, not full-budget search repetitions or evidence of a new history gain.

Selection sensitivity remains. With even-indexed panels used to select among the saved checkpoints, cell seeds 61/62 gain 17.13/13.58 points on the proposal subset but lose 2.68/8.54 points on its odd-indexed complement; seed 63 improves both. We retain all six task/seed rows and all 252 five-panel subset diagnostics. Because these checkpoints were originally selected on all ten panels, the complements are descriptive, not independent confirmation. A new selection study must choose checkpoint rounds and candidates without consulting the confirmation panels; robustness across new biological partitions remains unestablished.

5. What changed in the manuscript

Appendix E.7 reports 40 fresh fixed-D00 fits over ten laboratory identities and four endpoints, with scores independently reproduced. Laboratory 0 is the best on Norman and below the independent-laboratory mean on protein/VCC. This one-seed diagnostic does not rotate search controllers or address DTI.

Consolidated tables retain all original short/extended outcomes and trajectory diagnostics. All six or 24 slots were executed, so retrospective target crossings are not realized compute savings. Sections 4.7-4.8 and Appendix E distinguish collaborative access, history, allocation and validation evidence.

Response to Reviewer AfJB

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Thank you for identifying the relevant prior work and the need to separate model-development benefits from additional laboratory access. We have revised both the positioning and the experimental comparison rather than treating the original main table as evidence for the history mechanism.

1. Relation to existing collaborative research systems

The revised related-work section discusses all three systems explicitly. The Virtual Biotech coordinates specialist agents around biomedical evidence for therapeutic decisions. AutoScientists maintains shared experimental state, including unsuccessful directions, and coordinates subsequent experimentation. Helmsman is directly relevant to automated federated design: its planning, coding and simulation-based diagnosis already connect LLM agents to federated system construction and refinement. We have updated its citation to the ICLR 2026 publication.

BioCoLoop's contribution is the biological model-development interface and its controlled evaluation. Task adapters connect interaction, observed-response proteomic and perturbation predictors to the same local-fitting, development-evaluation and proposal interfaces. The coordinator distinguishes training-example-weighted parameter updates from equal-panel development evidence. Source-specific heads allow studies with different response objectives to contribute to a shared representation while the target development population remains explicit. The evaluation then separates laboratory participation, proposal-history conditioning and training allocation.

This is a narrower and more concrete contribution than inventing shared research memory or automated federated design. We now state the four-factor finite search space directly, and do not portray it as open-ended architecture or source-code discovery. The abstract, introduction and conclusion have been rewritten around the implemented framework and the evidence for each decision.

2. Does the history mechanism itself improve search?

We added random search without replacement and fixed GP-EI Bayesian optimization under the same ten-laboratory access as direct and history-guided LLM search. All methods use the same twelve configurations, development panels, initialization and six-proposal cap. The new main-text table reports all five endpoints; Appendix E provides exact random-search variability, shared-first-proposal and realized-fit-count controls.

The result is not a general history advantage. Qwen direct and history select equally scoring models on all five endpoints. Luna history improves Norman from 22.30 to 23.11 Top-1, versus a random expectation of 22.66, but decreases DTI from 94.60 to 94.27 AUROC. GP-EI matches direct Luna on DTI. Proteomics retains the starting configuration, and the VCC score attained by the LLM searches is already attained by the fixed configuration. These controls directly address the alternative explanation that conventional configuration search is sufficient in much of this small design space.

The comparison reuses verified complete candidate fits, with development-only selections sealed before additional scoring. It is a retrospective one-training-seed-per-endpoint study, not new multi-seed evidence. Three-seed source-defined laboratory controls are also reported separately: Qwen history is below direct on both proteomics and cell prediction, and the additional Luna feedback experiment yields only a small protein change and cell ties. We retain these results rather than extending the combined-system gain to unsupported component claims.

Appendix E.6 adds a separate three-seed mechanism study with 48 fresh fits over 16 designs, using four study-cell contexts from Replogle and Nadig and gene-disjoint train/development/test partitions. Direct and structured history each receive six requests, 16,940 optimizer steps and 532,840 training-example exposures per search. Raw and structured history see the same queried development facts. Held-out Top-1 is 22.625 for direct/raw, 22.888 for structured, 22.668 for random and 23.152 for GP-EI.

Structured-minus-direct is +0.263 points with paired gene-cluster 95% interval [-1.022, 1.568], conditional on the fitted, selected models. This changes search decisions but does not establish a stable independent history advantage or replace the original endpoints.

We also repaired a concrete feedback-transport omission in a new implementation version (Appendix E.8): the original source-defined adapters averaged ten panels before returning one worker result, losing the true worst-panel diagnostic while preserving the main score. Reconstructing all 72 source-defined checkpoints reproduces primary/loss exactly and restores 720 labelled panel scores. Eight paired ten-round CPU canaries preserve selected weights and training trajectories. These checks establish faithful transport, not an improvement in history-guided search. Complementary development-panel diagnostics still show selection reversals, and their reused checkpoints prevent interpreting them as independent confirmation.

3. What the revised empirical contribution supports

The original main table is now explicitly an access-scenario comparison, with a Labs column. The new same-access table addresses search quality. Re-evaluation and new development-roster diagnostics identify validation sensitivities associated with poor held-out generalization.

Appendix E.7 additionally reports 40 fresh fixed-design CPU fits across all ten single-laboratory identities on four endpoints. Laboratory 0 is the highest-scoring independent laboratory on Norman; identity variation therefore matters. This one-training-seed sensitivity does not rotate every controller, provide a DTI rotation, or equalize one- versus ten-laboratory optimizer steps.

We also completed 360 fresh fits to compare the allocation policy with uniform, ordinary score-only, learning-rate-stratified score-only and random promotion at identical training/evaluation counts. The original Norman gain over uniform is reproduced, but score-only promotion recovers most of it (21.70 versus 22.32), and the exact random-promotion expectation is 22.63. The trajectory policy improves Tahoe relative to both alternatives, while random is better on VCC. We report these mixed controls in Section 4.4 and Appendix E.5 rather than claiming a generally superior new allocation algorithm. The revised contribution is the implemented biological interface and its controlled empirical assessment, with data-access, search and allocation effects distinguished.

Response to Reviewer cR5g

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Thank you for the concrete concerns about conventional tuning, laboratory heterogeneity and the interpretation of the baseline comparison. We have added the missing controls and reorganized the evidence around those questions.

1. Conventional hyperparameter optimization

The revision adds a same-access table comparing random search without replacement, fixed GP-EI Bayesian optimization, direct LLM search and history-guided LLM search. All use ten laboratories, identical development panels, the same twelve-design library and a six-proposal cap after the initial configuration. Exact enumeration of 462 candidate subsets reports random search's expected outcome and search-policy variability; it does not count these subsets as independent training seeds.

For example, DTI AUROC is 94.60 for GP-EI, 94.45 in expectation for random search and 94.27 for history-guided search. On Norman, Luna history reaches 23.11 Top-1, versus 22.30 for GP-EI/direct and 22.66 for random search. Other endpoints include ties and cases where the fixed configuration is already competitive. Thus the new evidence supports task-dependent search behavior, not general superiority to conventional tuning. We also report controls matching the actual number of valid fits when an LLM proposal fails.

A separate three-seed, gene-disjoint study adds 48 fresh fits over 16 designs and four Replogle/Nadig study-cell contexts (Appendix E.6). With six requests and matched fitting counts, structured history reaches 22.888 Top-1, direct/raw 22.625, random 22.668 and GP-EI 23.152. Structured-minus-direct is +0.263 points, with a conditional paired gene-cluster 95% interval [-1.022, 1.568]. We report every control and do not claim that this study establishes superiority to ordinary tuning.

2. Laboratories defined by different studies

The main ten-client experiment partitions one source dataset; it is not ten independently operated institutions. We now make this explicit and separate it from the source-defined laboratory experiments.

The proteomic experiment treats PTPC, Lin, Ruprecht and decryptE as four laboratories. Their different response objectives use source-specific heads and a shared representation; the same target development panels select checkpoints. In the fixed-design transfer comparison, adding the three external studies increases target AP from 34.41 to 37.53.

The cell control compares four same-source VCC groups with VCC, Replogle/K562, Nadig/HepG2 and Jiang/IFNB/K562 as four source-defined laboratories. Each arm has four laboratories and 60 training interventions, with the same target anchor. The external-source arm scores 28.82 Top-1 versus 30.70 for the same-source arm. Underlying cell counts and gene panels remain different and are documented. DTI likewise reports each auxiliary source: Davis/Human improve the fixed target model, while BioSNAP and the combined-source setting do not.

These experiments test real differences between released studies, although execution is simulated on one host. We do not present heterogeneous data as automatically beneficial. Appendix E also reports the completed three-seed direct/history controls within the source-defined protein and cell settings, including their negative results.

Appendix E.8 separately audits a repaired feedback payload for these source-defined adapters. Originally, ten target panels were averaged inside one returning worker, so the outer worst-client field incorrectly described that mean rather than the true worst panel. The repair retains panel labels, scores and dispersion without changing primary/loss selection. Reconstructing 72 saved checkpoints yields identical primary/loss and restores 720 panel scores; eight paired ten-round CPU canaries preserve training trajectories and selected weights. The published source-defined search results remain unchanged, and a search benefit from this repair has not yet been measured.

3. Why direct search can be worse than the fixed model

“Direct” denotes feedback-free LLM configuration search, not training one model once. Every method ultimately deploys one selected predictor; ten laboratories are data holders contributing to fitting, not a ten-model ensemble. Both direct and history search retain the model with the best development score, whereas “fixed” is a single predefined configuration. Retaining a better development model therefore need not retain a better test model.

We investigated this failure rather than changing the reported test selection. Reconstructed VCC/Tahoe predictions show that equal-laboratory and pooled-intervention macro aggregation agree exactly. Small gains on 20 VCC or 60 Tahoe development interventions can accompany much larger test decreases. Re-evaluation under 20 new paired development-option rosters further shows that Qwen/VCC and Tahoe's original development advantages are not stable to distractor/option-order changes. No models were retrained or reselected in this diagnostic.

The new source-defined panel diagnostic also retains mixed outcomes. Selecting among saved checkpoints on five fixed panels improves cell seeds 61/62 by 17.13/13.58 points on those panels but decreases the complementary five by 2.68/8.54 points. The checkpoints were already selected using all ten panels, so this is descriptive evidence of sensitivity, not independent confirmation or a new biological split. The larger gene-disjoint study improves all terminal searches over their fitted D00 references but still includes an interim and a context-specific development/test reversal; validation error is not eliminated.

4. Scope and presentation of the contribution

The original access comparison now has an explicit Labs column and is no longer used to attribute the benefit of additional data to the LLM loop. The same-access search table addresses that attribution directly. We describe the method as a common biological model-development interface with four editable configuration factors, distinct from open-ended model invention. Related work now discusses Virtual Biotech, AutoScientists and Helmsman. A separate 360-fit allocation comparison adds ordinary score-only and random promotion: it reproduces the original benefit over uniform allocation but shows that much of it is also attained by simpler policies. The revised conclusion states these results separately from collaborative fitting and proposal history.

We additionally fitted fixed D00 independently at every laboratory identity on four endpoints: 40 fresh CPU fits at one seed (Appendix E.7). Laboratory 0 is the best on Norman and below the independent-laboratory mean on protein/VCC. We therefore report the full identity distribution separately from the original multi-seed workflow comparison. Full controller rotations, the DTI rotation and one-laboratory training matched to ten-laboratory optimizer steps remain outstanding; equal communication rounds alone do not equalize computation.